

Date	15 March 2026
Team ID	XXXXXX
Project Name	VBCUA — Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

SAMPLE PROJECT DOCUMENTATION

PROJECT OVERVIEW

Project Title	Voice-Based Concept Understanding Analyser (VBCUA)
Project Description	VBCUA leverages localized Speech-to-Text and semantic analysis models alongside cloud-based Large Language Models to offer an automated evaluation workflow for verbal technical explanations [cite: uploaded:README.md]. The application provides computer science students and technical trainees with instantaneous scoring arrays split between semantic alignment matching and auditory fluency pacing metrics [cite: uploaded:config.py].

CORE MODULE SPECIFICATIONS

1. Audio Transcription	Utilizes the OpenAI Whisper framework model size to translate uploaded audio explanation streams (.mp3, .wav, .m4a formats) into raw text strings cleanly, isolating signal data configurations automatically.
2. Semantic Matcher	Powers structural conceptual analysis via Sentence-Transformers (all-MiniLM-L6-v2) vector mapping [cite: uploaded:README.md]. It cross-references user verbal transcription strings against verified baseline topic text models to output a semantic similarity metric holding 60% of the overall evaluation weight.
3. Fluency Analyzer	Tracks acoustic audio features using Librosa and NumPy computations [cite: uploaded:README.md]. The subsystem processes raw signal attributes to extract phrase gap frequencies, pause counts, runtime speech durations, and maps them to a words-per-minute (WPM) speed baseline accounting for 40% of the scoring matrix.
4. AI Critique Generator	Hooks securely into the external Google Gemini API (gemini-1.5-flash) to render diagnostic qualitative feedback cards pinpointing specific omitted terms and conceptual progression tracking charts [cite: uploaded:config.py, uploaded:README.md].

OPERATIONAL SCENARIOS

Scenario 1 (Standard Flow)	A student uploads a 45-second explanation regarding the "Fractional Knapsack problem". The Whisper pipeline converts it to clear text arrays, Sentence-Transformers measures context alignment with baseline greedy-algorithm definitions at 85%, Librosa verifies a speech rate of 135 WPM, and the Gemini API prints clear grading summaries alongside a local SQLite save entry.
Scenario 2 (Network Offline Flow)	A technical trainee activates an evaluation during network latency drops. The local machine learning calculations (Whisper and Sentence-BERT vector matching) execute normally on the system host. When cloud connectivity timeouts block the external Gemini model request, the internal fallback engine engages immediately to provide localized diagnostic breakdowns seamlessly with zero pipeline failure.

CONCLUSION & OPERATIONAL VALUE

By establishing automated criteria splitting semantic verification (60%) and pacing speech attributes (40%), VBCUA completely changes traditional rote multiple-choice checking patterns [cite: uploaded:config.py]. It replaces unstandardized mock reviews with cost-effective, local standalone infrastructure, ensuring engineering students can practice verbal articulation competencies fluently and with complete technical accuracy [cite: uploaded:README.md].